

Inequations

Inequations, also known as inequalities, represent relationships between two quantities where they are not necessarily equal. They use symbols such as $>$, $<$, $\geq$, and $\leq$ to indicate whether one quantity is greater than, less than, greater than or equal to, or less than or equal to the other. Inequations can involve linear expressions, indicating a range of possible solutions, or more complex forms like absolute value inequalities, requiring consideration of multiple cases to capture all possible solutions. When solving inequalities, the focus is on isolating the variable while maintaining the relationship denoted by the inequality symbol, with special care taken when multiplying or dividing by negative numbers. Inequations are key in algebra, calculus, and optimization, providing a way to describe conditions where equality does not necessarily hold.

Example 3: Solve $(x \cdot (x - 1)) < \frac{(x-1) \cdot (x+1)}{2}$

To solve the inequality $(x \cdot (x - 1)) < \frac{(x-1) \cdot (x+1)}{2}$, let's go through the steps:
Expand the inequality:

$$x^2 - x < \frac{x^2 - 1}{2}$$

Multiply both sides by 2 to clear the fraction:

$$2x^2 - 2x < x^2 - 1$$

Rearrange terms:

$$2x^2 - x^2 - 2x + 1 < 0$$

$$x^2 - 2x + 1 < 0$$

Although the expression becomes zero when $x = 1$, the quadratic $x^2 - 2x + 1$ is non-negative for all real values of x , indicating it is never less than zero.

$$\frac{-\infty \quad \dots \quad 1 \quad \dots \quad +\infty}{> 0 \quad \quad = 0 \quad \quad > 0}$$

Since the square of any real number is non-negative and cannot be less than zero, there are no real values of x that satisfy the inequality. Therefore, the solution to the inequality $(x \cdot (x - 1)) < \frac{(x-1) \cdot (x+1)}{2}$ is the empty set, denoted as $\emptyset$. This means there are no real values of x that make the inequality true.

Example 4: Solve $\frac{(2x+1)^2}{3} \geq x + 2$

To solve the inequality

$$\frac{(2x+1)^2}{3} \geq x + 2$$

Expanding the left side:

$$\frac{4x^2 + 4x + 1}{3} \geq x + 2$$

Multiply both sides by 3 to clear the fraction:

$$4x^2 + 4x + 1 \geq 3x + 6$$

Rearranging terms:

$$4x^2 + x - 5 \geq 0$$

To find the solutions to this quadratic inequality, is possible to use the quadratic formula or by factoring the quadratic expression, now it will be done using the quadratic formula:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

For $a = 4$, $b = 1$, and $c = -5$, we have:

$$x = \frac{-1 \pm \sqrt{1 + 80}}{8}$$

$$x = \frac{-1 \pm \sqrt{81}}{8}$$

$$x = \frac{-1 \pm 9}{8}$$

So, the solutions are $x = -\frac{5}{4}$ and $x = 1$.

Is necessary to test intervals in the critical points $x = 1$ and $x = -\frac{5}{4}$.

When $x < -\frac{5}{4}$, the expression is positive.

When $-\frac{5}{4} < x < 1$, the expression is negative.

When $x > 1$, the expression is positive.

$-\infty$	$\dots$	$-\frac{5}{4}$	$\dots$	1	$\dots$	$+\infty$
> 0		$= 0$	< 0	$= 0$	> 0	
$\uparrow$					$\uparrow$	

Since we want the expression to be greater than or equal to zero, the solution is:

$$x \in (-\infty, -\frac{5}{4}] \cup [1, +\infty).$$